Supplemental File ST3. ONT-based shotgun metagenomic sequencing reads assigned to microbial taxa in YNP samples collected using different filtration methods. Only taxa with cumulative read counts exceeding 1,000 are included.

Species	Domain	YNP01	YNP02	YNP03	YNP04	YNP05	YNP06	YNP07	YNP08	YNP09	YNP40	YNP41	YNP42	YNP46	YNP47	YNP48	YNP55	YNP56	YNP57	YNP61	YNP62	YNP63	YNP64	YNP65	YNP66	YNP67	YNP68	YNP69	YNP70	YNP71	YNP72	YNP73	YNP74	YNP76	YNP77	YNP78
		FSS3 (Sterivex)			FSS1 (Sterivex)			FSSe2 (Sterivex)			Sterivex control			OSe (Sterivex)			OS (Sterivex)			FSS3 (PC)		FSS3 (Supor)		FSS1 (PC)		FSS1 (Supor)		FSSe2 (PC)		FSSe2 (Supor)		Ose (PC)		OSe (Supor)	PC	Supor
Pyrobaculum sp. WP30	Archaea	211	125	137	70	81	81	185	143	130				50	52	18	723	543	851	85	480	577	169	290	334	313	325	449	105	358	677	32	58	461		
Pyrobaculum sp. OCT_11	Archaea	10	12	17	2	7	8	18	17	13				5	7	1	55	52	82	5	49	54	9	25	20	28	21	25	3	33	45	2	4	41		
Thermoproteus sp. CP80	Archaea	33	12	13	14	7	13	25	27	30				6	6	1	43	31	62	7	25	48	19	37	26	32	17	44	6	45	59	2		30		
Pyrobaculum sp. JCHS_4	Archaea	15	7	12	8	4	2	12	11	5				4	3		42	27	33	3	29	46	7	15	15	24	16	35	5	26	52	2	3	27		
Pyrobaculum ferrireducens	Archaea	19	4	6	5	2	7	9	8	5				2	2		39	24	36	3	23	21	5	15	12	11	11	22	2	21	35		2	21		
Thermoprotei archaeon	Archaea	18	13	16	9	14	12	15	16	16				7	8	2	10	18	17	3	6	29	4	27	20	13	25	18	2	29	46	6	7	3		
Desulfurococcaceae archaeon AG1	Archaea	33	3	11	8	11	4	22	16	12				4	8		9	8	7	5	6	16	4	8	9	10	9	6	1	20	40	1	4	5		
Pyrobaculum aerophilum	Archaea		1			1	2	3	2	2				1	1	1	3	5	9	1	2	4	3	1	2	2	1	5	4	2	5			4		
Thermoproteus sp. JCHS_4	Archaea	2	1	2		2	1	2		3				1			9	8	7	2	6	9	1	4	7	3	5	4	2	4	10			3		
archaeon HR02	Archaea	6		6	3	6	2	10	6	11				12	13	4	9	9	16	3	7	9	7	22	19	14	12	29	4	17	12	11	5	6		
Caldivirga sp. MU80	Archaea	4	1	4	1	6	3	6	4	4							7	7	8		5	10	2	6	7	7	7	13	1	2	11		1	1		
Candidatus Caldiarchaeum subterraneum	Archaea	2				1		10	9	16				89	151	60	2	2	4		2	7	2	13	9	3	11	36	6	10	14	94	109	3	1	
Pyrobaculum islandicum	Archaea		3	2	2	3	3	2		1				1			3	3	7	1	2	9	4	3	6	5	4	6	1	2	7			1		
Thermocrinis ruber	Bacteria	17412	8538	9696	8923	8028	9651	5090	4288	4855				4108	4923	2466	67832	54675	73802	9844	71470	68397	14041	47491	47187	53314	54867	33795	4234	23731	35438	4982	5825	68063		5
Synechococcus sp. JA-2-3B'a(2-13)	Bacteria							506	468	368				19023	9508	7973	9	4	8	1	1	14		35	75	122	152	5885	809	1029	2034	6944	9200	6		1
Thermus aquaticus	Bacteria			3	4	3	2	21	17	13				289	84	178	12	19	17	8	20	26	14	31	63	35	38	588	249	54	233	146	235	22		
bacterium HR17	Bacteria	54	30	31	42	41	53	47	30	34		1		111	111	58	422	372	480	312	276	1417	347	2072	2422	502	593	1175	210	225	395	127	128	338	1	
Clostridioides difficile	Bacteria															1										8	3						1			
Candidatus Thermochlorobacter aerophilum	Bacteria						1	139	122	134				393	315	192		2			1	59		1	91	32	31	188	44	3213	4799	1732	1986	3	1	2
Thermocrinis sp. GBS	Bacteria	173	95	121	99	94	124	48	37	53				43	49	24	682	519	751	94	684	682	147	503	506	567	583	362	48	283	417	48	65	665		
Armatimonadetes bacterium CP1_7O	Bacteria			1	1		1	8	9	8				307	98	143	2	1	4	3	2	14	8	22	30	8	12	234	83	70	208	156	266	1		
bacterium HR21	Bacteria			4	2			42	26	29				813	454	521	6	7	6	1	9	15	6	16	31	42	49	332	72	131	327	1194	1852	11		
Armatimonadetes bacterium DC	Bacteria		1					11	6	8				207	84	105	1	2	2	2	3	8	2	17	30	5	11	181	43	51	199	138	247	3		1
Synechococcus sp. JA-3-3Ab	Bacteria							36	40	28				1269	637	521					2	5			3	5	8	426	74	107	295	365	510			
Bacteroidetes bacterium	Bacteria			2	6	2	4	177	146	133				500	408	228	3	1	3	5	1	13		17	23	20	20	456	74	441	757	497	617	2		9
Thermomicrobium roseum	Bacteria							13	10	16				383	246	283	2	2	3	1		4	1	3	3	3	14	190	44	12	39	243	462	1		
Armatimonadetes bacterium	Bacteria				1			4	8	8				136	40	79			3	5	4	5	2	6	29	4	6	149	55	40	109	91	149	6		1
Roseiflexus sp. RS-1	Bacteria							17	13	16				593	275	195	1		1			3		1	9	3	3	67	10	153	270	47	62			
bacterium HR16	Bacteria		2	1	2	3		8	5	17				430	375	233	11	6	6	5	5	34	5	37	42	17	20	165	32	140	261	374	517	5		
Synechococcus sp. 65AY6A5	Bacteria							28	25	25				898	448	393						1		1	2	3	10	332	52	113	205	308	406		1	
Thermoflexus hugenholtzii	Bacteria						2	2		2				37	8	10	1		1		2	5	1	4	3	4	5	15	2	5	6	22	34	1		
Gloeomargarita lithophora	Bacteria							38	32	34				205	153	87					1	45		6	57	2	4	195	34	2183	3190	92	122	1		5
Ralstonia solanacearum	Bacteria								1					1	1	2																2	2			
Lactobacillus rhamnosus	Bacteria																																			
Synechococcus sp. 63AY4M2	Bacteria							28	25	20				888	469	381	1	1				2			1	2	8	324	44	79	163	286	370		1	
bacterium HR15	Bacteria							2	3	6				182	120	78	1	2		3	4	8	1	11	13	11	11	148	30	38	94	160	259	2		
Chloracidobacterium thermophilum	Bacteria		1	2		1	1	17	23	10				16	8	10	1	3	3		3	15		2	12	2	3	205	27	425	699	5	13	6		
Acidobacteria bacterium	Bacteria		1	2	2		3	67	45	65				158	133	79	8	8	19	3	15	20	4	28	19	13	18	232	35	271	360	503	534	10		61
Synechococcus sp. 65AY6Li	Bacteria							9	6	9				367	194	163									3		2	140	21	39	98	142	163			
Armatimonadetes bacterium JP3_11	Bacteria							1	1	2				54	20	29	2				1			7	8	4	2	56	17	16	57	41	66	2		
Thermus scotoductus	Bacteria							6	3	4				73	49	45	9	8	5		1	15	1	5	11	6	5	89	24	16	51	40	72	11		
Stenotrophomonas maltophilia	Bacteria										1	1		1																				2	244
Lactobacillus mudanjiangensis	Bacteria																																			
Hydrogenobacter hydrogenophilus	Bacteria	22	12	8	3	1	1	18	11	12				25	39	5	200	207	235	18	238	54	7	18	10	10	9	56	2	42	57	34	31	268		
Chloroflexus sp. Y-396-1	Bacteria							12	10	4				101	63	46		1	2			1			6	3	2	46	11	87	165	119	135			
Fischerella thermalis	Bacteria							3		2				57	30	27	2			1				4	4	2	5	33	10	17	45	37	55			
Chloroflexus sp. MS-G	Bacteria							12	2	10				55	43	36		1				1			3	1	2	52	15	87	157	101	108			
Aquificae bacterium	Bacteria	7	5	1			1	2	1					3	2		11	8	9	1	7	16	1	4	8	5	6	16	1	12	16	6	4	10		
Chloroflexi bacterium	Bacteria	1						3	4	2				59	36	19	4	1	8	3	6	7	1	8	9	2	6	20	5	10	40	37	59	9		11
Sulfurihydrogenibium yellowstonense	Bacteria	37	11	14	9	4	6	8	4	7				12	21	12	69	63	72	12	69	103	22	43	40	46	66	69	1	59	69	16	20	80		
Sulfurihydrogenibium sp. YO3AOP1	Bacteria	7	2	1	4		3	11	3	6				3	12	2	4	6	16	11	9	40	12	7	8	8	9	34	7	23	33	8	11	12		
Candidatus Thermofonsia Clade 1 bacterium	Bacteria							4	4	8				1	2							2						32	10	9	19	16	43			1
Thermodesulfobacterium geofontis	Bacteria	37	11	20	27	17	35	4	12	7				13	26	11	45	56	66	4	48	100	21	125	113	112	123	31	1	67	56	13	8	67		
Meiothermus cerbereus	Bacteria							6		5				8	12	7		1				4		10	9		1	35	11	22	54	15	27			
Synechococcus sp. 63AY4M1	Bacteria							2	2	3				82	30	40					1	2				1		38	3	14	33	28	33			
Hydrogenivirga sp. 128-5-R1-1	Bacteria	1	1	2	2	5	2	1	3					2	3	2	3	4	3	3	1	12	5	15	14	14	17	6		10	12	1	2	1		
Campylobacter coli	Bacteria																																			
Caldilinea aerophila	Bacteria							6	5	4				47	33	21											1	29	8	12	28	21	26			
bacterium HR19	Bacteria	22	10	9	2	5	6	11	8	21				74	103	45	43	47	58	17	35	77	15	74	72	39	31	39		81	79	47	47	41		
Thermus sediminis	Bacteria				1			1	1	1				10		7		1			1	1		1	1		1	11	6	2	4	3	7	1		
Persephonella marina	Bacteria	2	4	1	1	1	3	2						1	1		10	4	5	5	7	17	2	19	18	29	24	11	1	10	14		1	2		
Clostridium sp. C105KSO15	Bacteria																																			
Hydrogenobacter thermophilus	Bacteria	5	6	2	6	4	4	3	2	1				9	8	5	17	7	11	2	6	16	3	24	16	8	15	12	4	9	15	7	10	2		
Chlorobium sp. 445	Bacteria							7	5	5				15	16	7						3			1	2	1	11		110	147	62	70			
Thermocrinis albus	Bacteria	1	4	3	1	5	1	1	4					1	6		14	16	27	6	18	33	12	23	26	26	31	17		14	19	4	2	27		